Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To reduction of 80% Dynol overspray concern on export Scorpio Model "

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**25.03.18**	**5.04.18**
Completion Date	**20.10.18**	**8.11.18**

Current Status: **CLOSED**

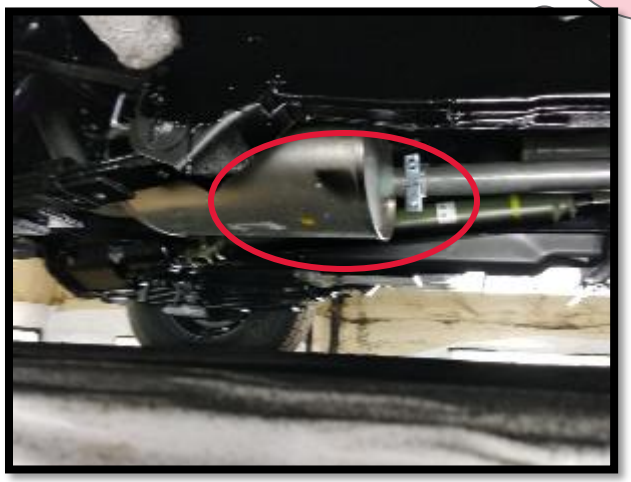
Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Sudhir Dhanokar	Manager	Production	9049992381	Dhanokar.sudhir@Mahindra.com
2	**Mentor**	Rahul Narkar	DGM	Manufacturing	9881080299	NARKAR.RAHUL@Mahindra.com
3	**Member 1**	Rajesh Dhanokar	Manager	Industrial Engineer	9881250515	Dhanokar.rajesh@Mahindra.com
4	**Member 2**	Praveen Patil	Manager	Process	8888897850	Patil.pravin@Mahindra.com
5	**Member 3**	Pankaj Tandle	Deputy Manager	Production	9922944452	Tandale.pankaj@Mahindra.com
6	**Member 4**	Chetan waje	Deputy Manager	Production	9028176386	Waje.chetan@Mahindra.com
7	**Member 5**	Nitin Deshmukh	Deputy Manager	PE	9922944521	Deshmukh.Nitin@Mahindra.com
8	**Member 6**	Danish Shaikh	Deputy Manager	CQA	8308844828	Shaikh.Danish@Mahindra.com

Step1. Select Subject

- Brief Description of the Problem:

Dynol Overspray on Exhaust , Body



- During Dynol application Dynol overspray on Exhaust , body , Cargo.

Step1. Select Subject

▪Brief Description : 80% Reduction the concern of Dynol overspray in Export vehicle

What : Dynol marks on body/ cargo

Where : Multimodel Shop , CAI yard .

When : Since Feb 2017.

Why: Chronic Repeat Concern, Production loss, Quality loss, Cost loss, Deliver loss.

Who: By CQA inspector.

How: 94% vehicle affected.

2. Justify the choice (reasons)

Mahindra

PROMISE 2019
Automotive Business

We will…

Double our revenue to ₹53,000 cr (USD 8 bn), while delivering superior financial performance

- Be the 2nd largest OEM in India by volume
- Double the overseas volume with significant presence in 10 markets

By being the most customer centric company

Brand and Customer Satisfaction
- Be among the Top 2 in Customer Satisfaction
- Be among the Top 3 in brand ranking in PV and #1 in UV & SCV
- Have 10 brands with CaPS > 50

Product and Technology
- Launch 18 products
- Be a pioneer in digital customer experience
- Develop one new disruptive business model

Crusade 2.0
- Drive brand preference through Crusade 2.0 of delivering ETR, FTR and DCE

People
- Be the #1 Great Place to Work in the manufacturing sector

Environment
- Reduce carbon and water footprint by 25%

Thus enabling our customers and stakeholders to Rise, as we progress towards being a Top 10 Global Automotive brand.

Mahindra

PROMISE 2019
Nashik Plant 1, Auto Division

We will…

- Produce 230k vehicles in a year as per market requirement
- Fulfill 100% IO requirement for doubling the overseas volume
- Maintain cost leadership in Auto Sector by reducing revenue cost of 5% YoY
- Improve IQS for Nashik models by 25%, ranking in top 3
- Achieve CaPS > 50 for all Nashik models
- Launch 7 new products on time
- Achieve defined Crusade targets through FTR, ETR and DCE
- Achieve operational excelle[illegible] through digitization
- Achieve No.1 status for en[illegible]ee engagement in the Auto Industry
- Make officers future ready [illegible]plement new technologies coming in the next 5-10 y[illegible]
- Reduce carbon & water foo[illegible]nt by 25%

Departmental Objective –

RTR Improve, BIQ of Dynol improve, Flashaves FRC Improve.

2. Justify the choice

Cost of Poor Quality

Material cost :13920 Rs/-

Cost of 1kg cloth = 58 Rs per kg
Avg 5kg required per week :- 58*5=290
Per month=290*4=1160

Cloth cost in one year
=1 month cloth cost x 12
=1160 x 12
=13920 Rs / Year

Total 13920 rs/year cloth consumption.

Labour cost :36232 Rs /-

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	18
Total Required Rework time per day	Min	180
Effective working time	Min	465
Operator required /day	nos	0.39
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	125.81
Avg no. of working days	days	24
Operator cost /month	Rs	3019.35
Part reject/month	Nos	0.00
Part Cost	Rs	0.00
Operator Cost/year	**Rs**	**36232.26**

Total Cost/ year = Material cost + Manpower cost = 13,920+36,232 = 50,152/-

2. Justify the choice (reasons)

1. RFI Concern reported
2. Field complaint reported

concern mail from RFI auditor

Delete | Respond | Quick Steps | Move | Tags | Editing

Mon 5/21/2018 10:04 PM

RH-CQA-NSK

Scorpio EXPORT RFI Report for 21 May 18

To: SCORPIO NOVAC

From: RH-CQA-NSK
Sent: Monday, May 14, 2018 10:05 PM
To: SCORPIO NOVAC
Subject: Scorpio EXPORT RFI Report for 14-May-18

Dear All,
Please Find SCORPIO EXPORT RFI Report for 14 May 18

SL	VIN NO	VARIANT	CONCERN DESCRIPTION	PART DESCRIPTION	F / V	DEFECT DESCRIPTION	SEVERITY	DEPARTMENT	Incidence Date
1	46790	SC	Crack	Reflector RR Bumper LH	V	Crack Reflector RR Bumper LH	V2	SQA	14.05.2018
2	46790	SC	Over Spray	Dynol Tail Gate/ Exhaust	V	Over Spray Dynol Tail Gate / Exhaust	V2	ICD	14.05.2018
3	46700	DC	Spot Burr	Cargo Inside RR	V	Spot Burr Cargo Inside RR	V2	SQA	14.05.2018
4	46644	SC	Deformation	RH Side Lenses Reg Plat Lamp	V	Deformation RH Side Lenses Reg Plat Lamp	V2	SQA	14.05.2018

No of V1+ Defects : 00
No of V1 Defects : 00
No of V2 Defects : 04

concern mail from Field

A	B	C	D	E	F	G	H	I	J
Sr. No.	Plant	Veh Vin No.	Model	Colour	Destination	Plant Dispatch Da	Plant defect	Defect Desciption (Plant)	Attribution
13	Nashik	H2E47210	SCORPIO(D/C)	A3ARTCWHIT	QATAR	21-May-17	2	Dynol on LHS front bumper & rediator grill	STCF
16	Nashik	H2E48022	SCORPIO(D/C)	A3ARTCWHIT	SOUTH AFRICA	22-May-17	1	Dynol on back door	STCF

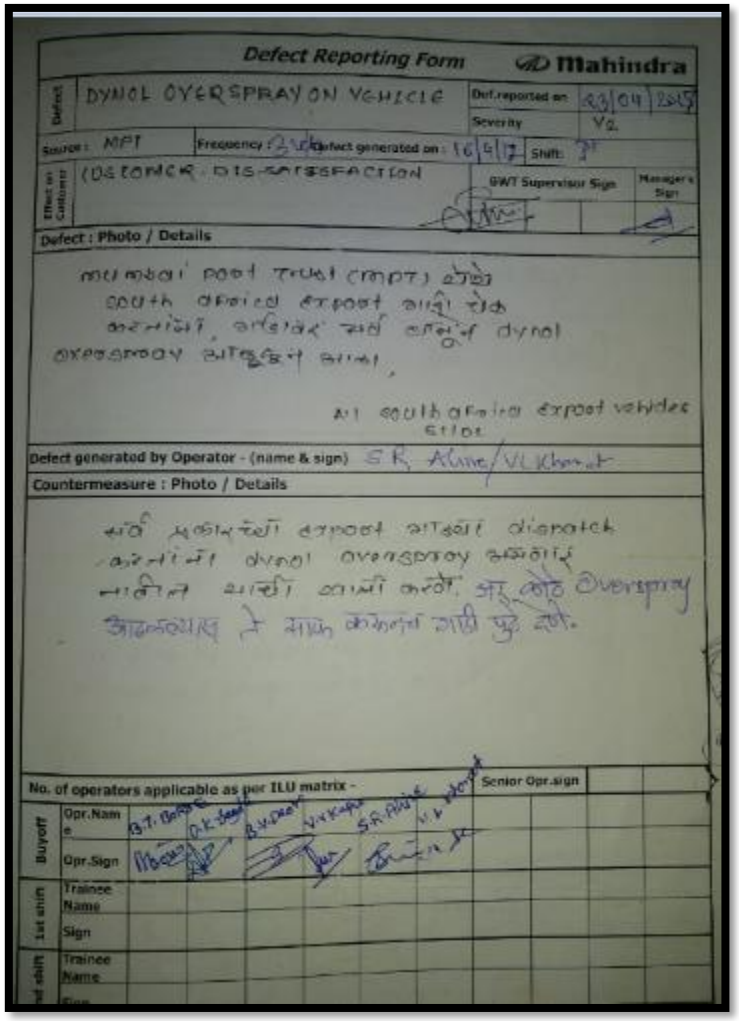
2. Justify the choice (reasons)

Problem Criticality (Severity)

- Dynol spray mark on body, Exhaust resulting in customer dissatisfaction due to bad appearance .
- Due to aesthetic bad appearance brand name affected in market .
- Vehicle flow increase due to Rework
- Require 15 to 20 mins for rework the one vehicle.

2.b Immediate Containment Action

- Vehicle reworked after Dynol application started.
- Defect communicated to the stage operator through DRF.



Defect Reporting Form — Mahindra

Defect: DYNOL OVERSPRAY ON VEHICLE — Def.reported on: 23/04/2017 — Severity: V2

Source: MPT — Frequency: 2 — Defect generated on: 16/5/17 — Shift: 3

Effect on Customer: CUSTOMER DISSATISFACTION — BWT Supervisor Sign — Manager's Sign

Defect : Photo / Details

Defect generated by Operator - (name & sign): S.R. Alune / V.L. Khorat

Countermeasure : Photo / Details

No. of operators applicable as per ILU matrix - Senior Opr.sign

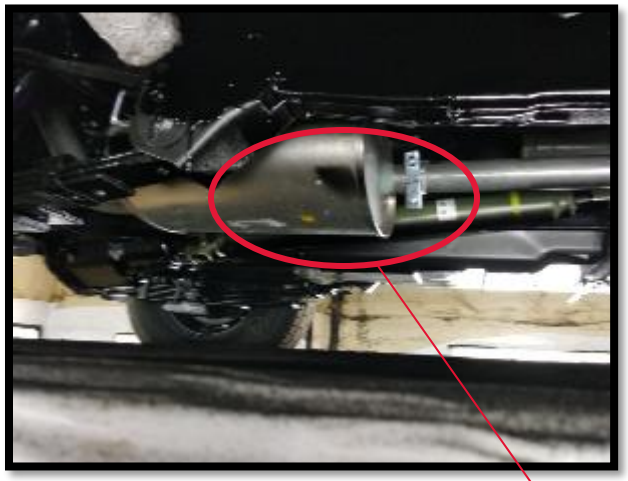
Buyoff — Opr.Name / Opr.Sign

1st shift — Trainee Name / Sign

2nd shift — Trainee Name / Sign

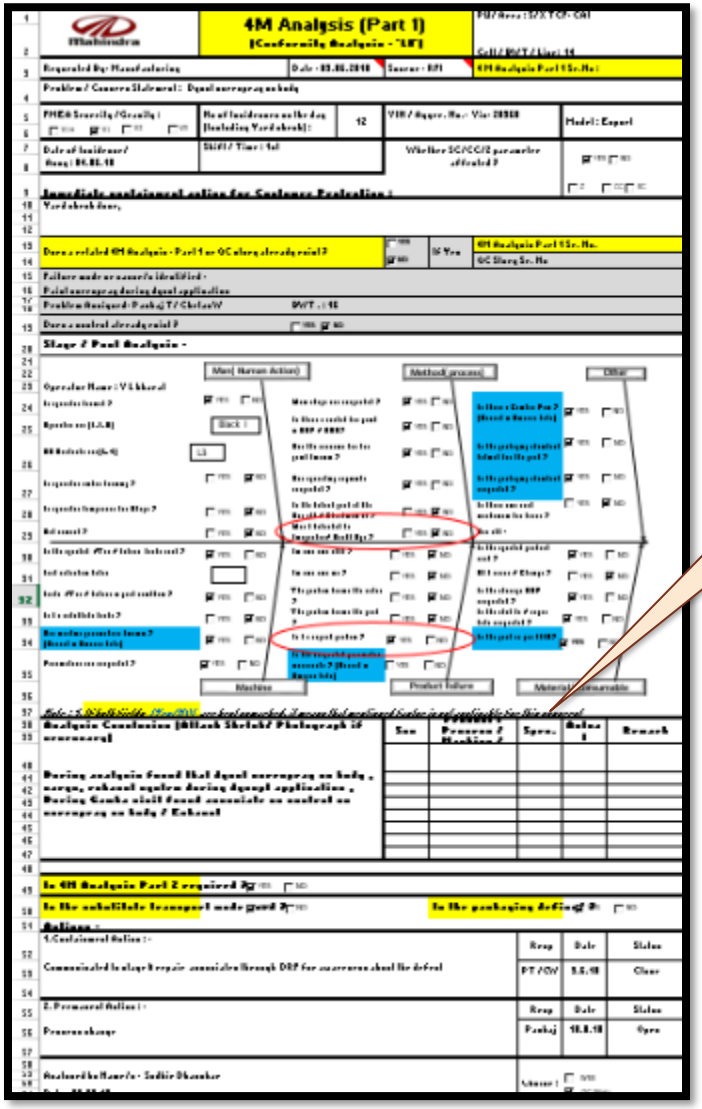
3. Understand the current situation

Dynol Spray marks on Exhaust , Body .



Dynol Overspray On Exhaust System / Body

Step 3. Understanding Current Situation



Inference from L0,
Concern is a repeat problem &
hence detailed analysis required

NO QUICK WIN.

4M analysis Good and Not so good observations	
MAN	Main step are respected by associate.
Method	Operator aware of key point and check not present after dynol application
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Process failure	SOS Followed
Material	Handling and shelf life of part adhered

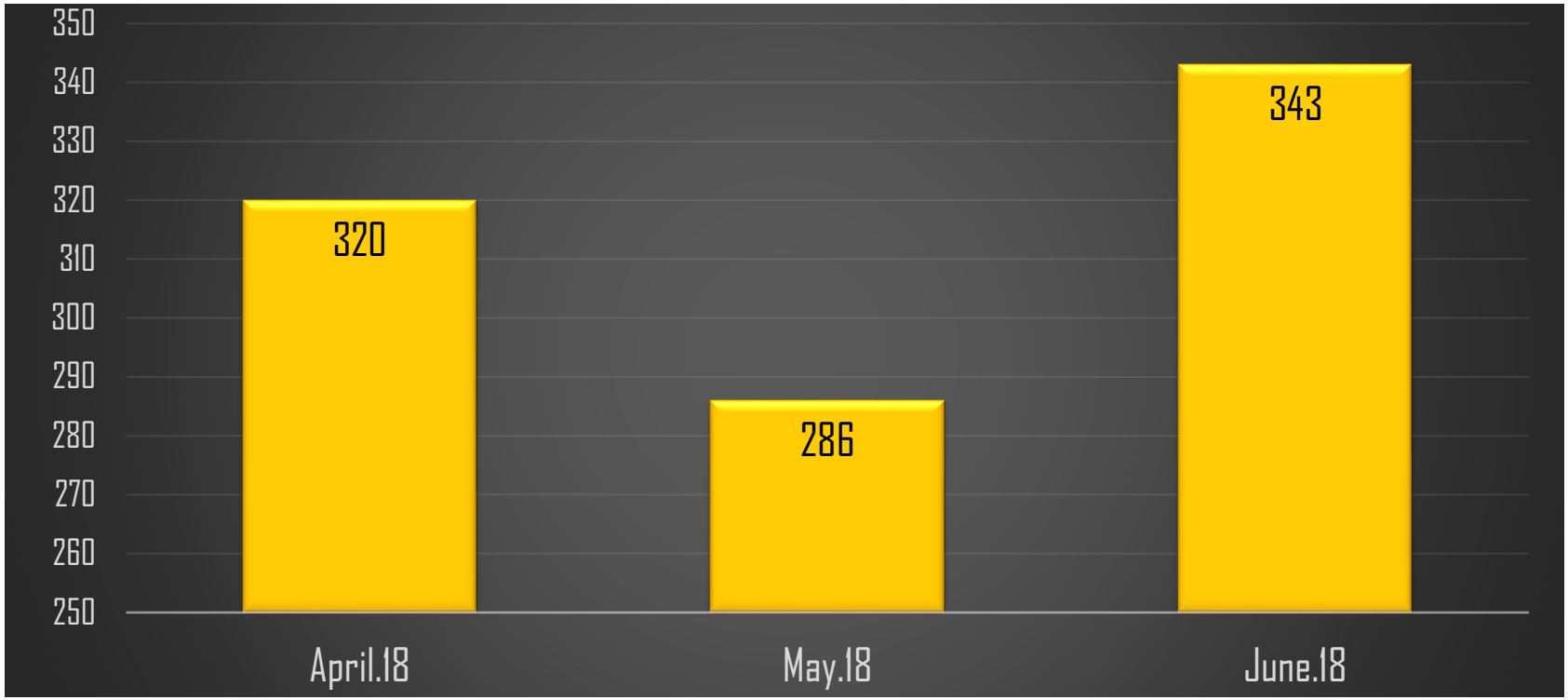
3. Understand the current situation

4M Analysis Observation.

- Concern Reported at Export RFI.
- **All L3/L4 Operators Working** on Stage.
- Concern Reported in **Export vehicle.**
- Concern reported as Dynol overspray at export RFI.
- **Rework** is done after RFI in yard area.
- **Rework** Takes more than **10 min** time to repair & checking.
- **RTR hampered** due to flow disturb.

3. Understand the current situation

Month wise **Dynol Oversparay** concern Trend at dynol for April-18 to June-18



INFERANCE:- **Defect incidences are consistent with monthly average of 316 inc/month.**

Step 3. Understanding Current Situation

Good & Not so good observations

- Operator following his SOS
- Dynol viscosity found ok
- Machine storage container temperature maintain.
- Issue reported in both the shifts.
- Issue reported at RFI / Field
- Not all the vehicles having this issue.
- Part need to clean after dynol application.
- Part getting rejected while rework.
- Not sure about 100% ok condition after rework
- Repair process is tricky .

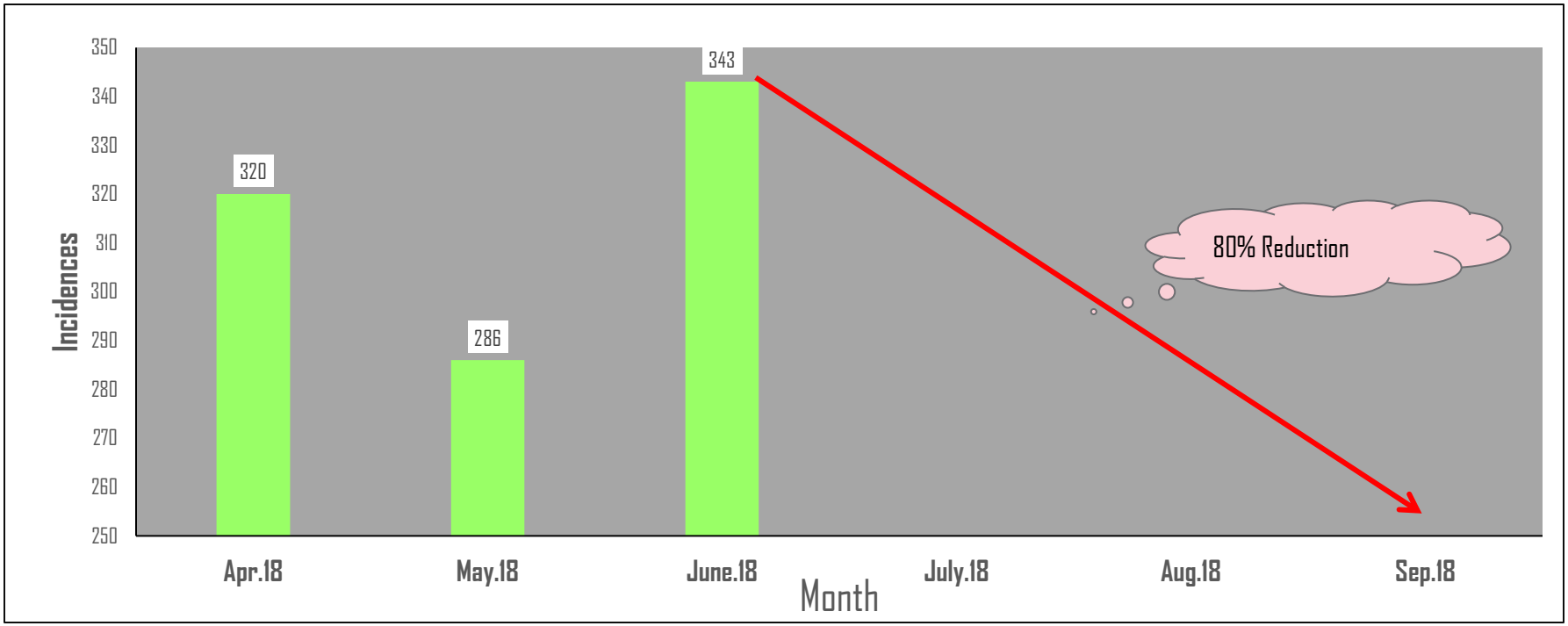
4. Select targets

Present Level:

Average 316 Defects/Month

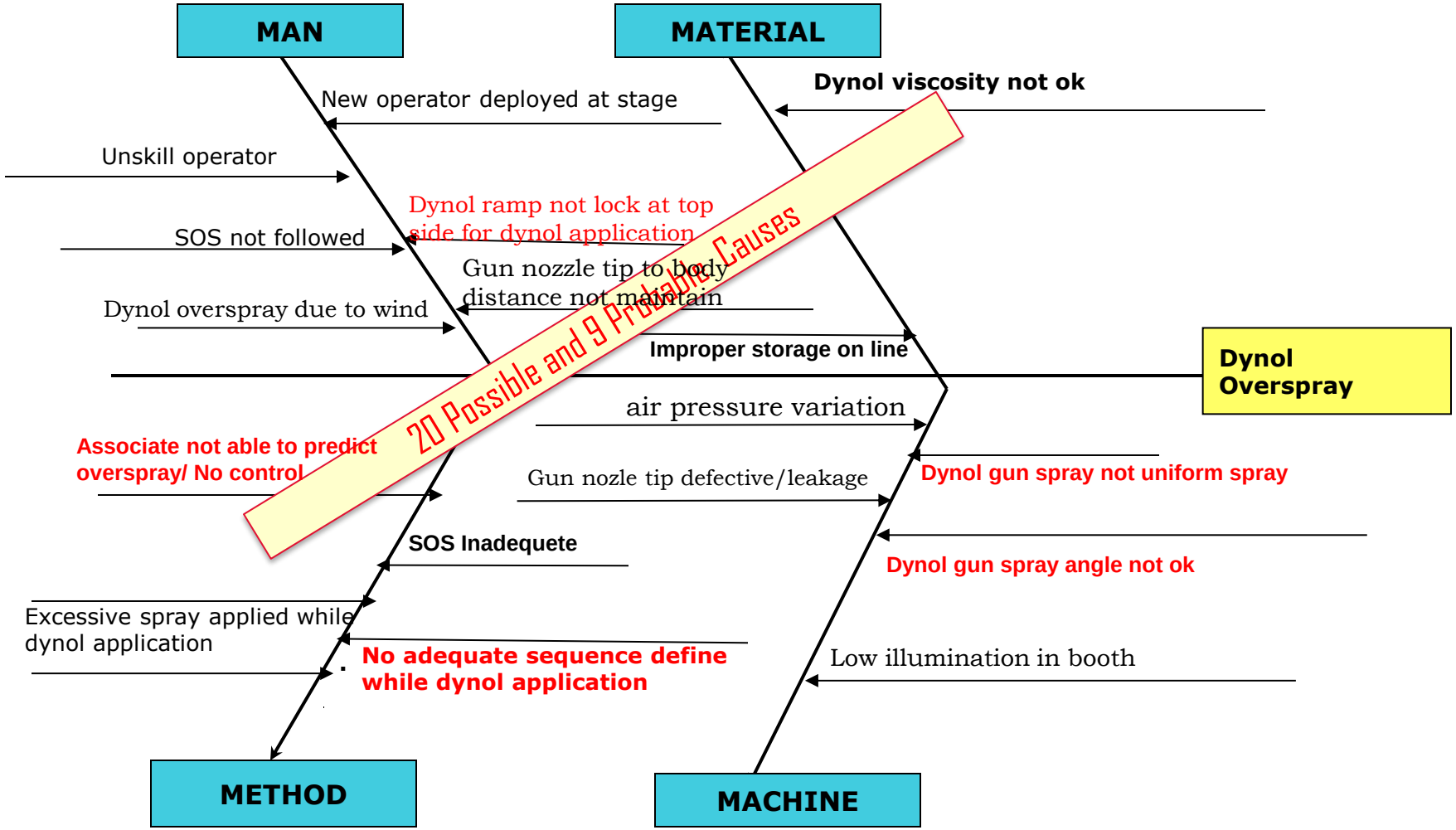
Target:

Reduction 80% of the defect Dynol overspray on body By Sept 2018



Step 5. Analysis

List Of Possible Causes



5. Analysis

5.1 List of Possible causes: Brainstorming

Sr. no	Possible Causes	Sudhir	Rajesh	Pravin	Sayali	Danish	Pankaj	Chetan	Total
1	New operator deployed at stage.	1	3	3	1	1	3	3	15
2	SOS Not Followed .	3	3	3	9	1	3	3	25
3	Unskilled operator	3	9	3	1	3	1	3	23
4	wrong tool used for spray	9	3	1	3	1	9	3	29
5	Improper storage on line	3	3	3	9	3	3	3	27
6	Associate not able to predict overspray/ No control	3	9	3	9	9	9	9	51
7	Dynol viscosity not ok	1	3	3	3	9	3	1	23
8	No adequate sequence define while dynol application	9	3	9	9	3	9	3	45
9	Gun nozle tip defective/leakage	1	3	3	9	1	3	3	23
10	Dynol gun spray angle not ok	3	1	9	9	3	9	9	43
11	Dynol gun spray not uniform spray	6	3	9	3	3	9	9	42
12	Dynol overspray due to wind	1	3	1	3	3	1	1	13
13	Dynol ramp not lock at top side for dynol application	3	9	3	9	3	9	9	45
14	air pressure variation	3	3	3	9	6	1	1	26
15	Low illumination in booth	3	3	3	3	3	3	3	21
16	Gun nozzle tip to body distance not maintain	1	1	1	1	3	1	3	11

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 5- Significant impact.
Threshold Value for selecting probable cause is 42.

Step 5. Analysis

5.1 List of Probable causes: Brainstorming

List of causes not considered as probable causes and reasons for not considering

Sr.No	Not considered Causes	Reasons
1	New operator deployed at stage.	Trained operator deploy on stage
2	SOS Not Followed .	Associate working as per SOS
3	Unskilled operator	All associate are L3 & L4 level
4	wrong tool used for spray	No other gun or tool available at stage
5	Improper storage on line	Storage instruction , checksheet & physical condition found ok
7	Dynol viscosity not ok	Dynol temperature condition maintain
9	Gun nozle tip defective/leakage	Tip no leakage or abnormality found
12	Dynol overspray due to wind	Dynol operation in booth so no atmosphere condition matter while dynol application
14	air pressure variation	Air pressure found ok
15	Low illumination in booth	Illumination check found ok
16	Gun nozzle tip to body distance not maintain	SOS follow found ok

5. Analysis

5.1 List of Probable causes: Brainstorming

SR NO	Possible Causes
6	Associate not able to predict overspray/ No control.
8	No adequate sequence define while dynol application
10	Dynol gun spray angle not ok
11	Dynol gun spray not uniform spray
13	Dynol ramp not lock at top side for dynol application

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Associate not able to predict overspray/ No control	Visual checking of Dynol application & SOS verification	20 cycle checked , stage associate not predict overspray from underbody / No control in process	Hypothesis Valid

SOS for Dynol Application	
SR NO	Observations
1	Found ok
2	overspray on bumper
3	Overspray on exhaust, door
4	Overspray on cargo,door
5	Found ok
6	Overspray on cargo
7	Found ok
8	Overspray on cargo
9	Found ok
10	Overspray on exhaust, door
11	Found ok
12	Overspray on cargo
13	Found ok
14	Overspray on exhaust, door
15	Found ok
16	overspray on bumper
17	Overspray on cargo
18	Found ok
19	overspray on bumper
20	Found ok

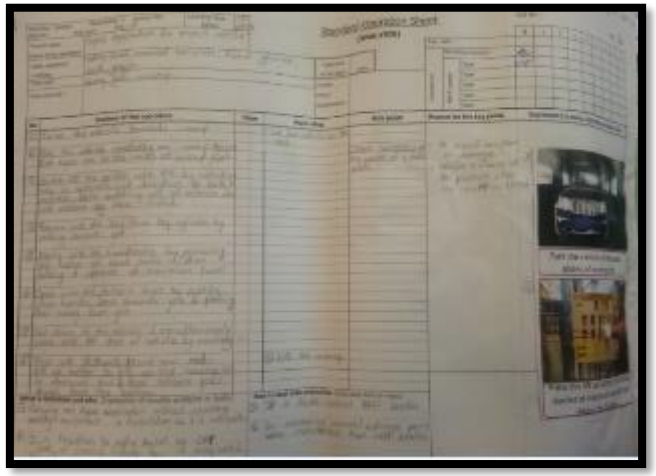
Dynol Overspray

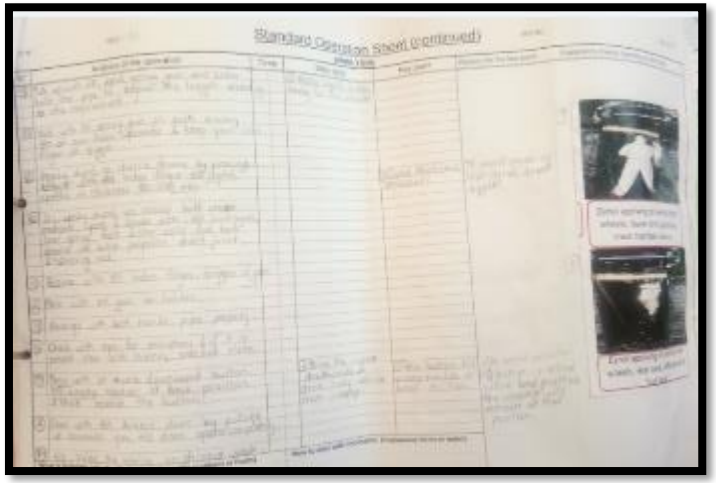
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	No adequate sequence define while dynol application	Visual checking of Dynol application & SOS verification	3 associate observed in both shift found variation in dynol application sequence, sequence not mentioned in SOS.	Hypothesis Valid

Dynol Application SOS





TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Dynol gun spray angle not ok	Visual checking	20 cycle observed , found at critical point dynol overspray on parts as spray angle excess observed, 11 vehicle concerns reported	Hypothesis Valid

Check Dynol Spray for Critical & Non critical Point		
SR NO	Critical Point	Non Critical Point
1	Overspray on exhaust	Found ok
2	Overspray on exhaust	Found ok
3	Overspray on exhaust front pipe, spare wheel	Found ok
4	Overspray on exhaust front pipe	Found ok
5	Overspray on exhaust pipe, front bumper	Found ok
6	Overspray on exhaust front pipe	Found ok
7	Overspray on exhaust front pipe, spare wheel	Found ok
8	Overspray on exhaust	Found ok
9	Overspray on exhaust	Found ok
10	Overspray on exhaust , Bumper	Found ok
11	Overspray on exhaust	Found ok
12	Overspray on exhaust , mudflap	Found ok
13	Overspray on exhaust	Found ok
14	Overspray on exhaust , mudflap	Found ok
15	Overspray on exhaust , cargo edge	Found ok
16	Overspray on exhaust	Found ok
17	Overspray on exhaust , Spare wheel	Found ok
18	Overspray on exhaust	Found ok
19	Overspray on exhaust , cargo edge	Found ok
20	Overspray on exhaust pipe, front bumper	Found ok

WIDER SPARY -520 needle size

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Dynol gun spray not uniform spray	Visual checking of Dynol spray at paintshop test bed	20 cycle observed for spray angle & spray uniform check at paintshop bed.	Hypothesis Invalid

SOS for tighnening sequence	
SR NO	Observations
1	Found ok
2	Found ok
3	Found ok
4	Found ok
5	Found ok
6	Found ok
7	Found ok
8	Found ok
9	Found ok
10	Found ok
11	Found ok
12	Found ok
13	Found ok
14	Found ok
15	Found ok
16	Found ok
17	Found ok
18	Found ok
19	Found ok
20	Found ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Dynol ramp not lock at top side for dynol application	Visual checking	20 cycle observed for ramp lock position while dynol application.	Hypothesis Invalid

SOS for tighnening sequence	
SR NO	Observations
1	Found ok
2	Found ok
3	Found ok
4	Found ok
5	Found ok
6	Found ok
7	Found ok
8	Found ok
9	Found ok
10	Found ok
11	Found ok
12	Found ok
13	Found ok
14	Found ok
15	Found ok
16	Found ok
17	Found ok
18	Found ok
19	Found ok
20	Found ok

Dynol Ramp position

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Associate not able to predict overspray/ No control	Visual checking of Dynol application & SOS verification	**Hypothesis valid**
2	No adequate sequence define while dynol application	Visual checking of Dynol application & SOS verification	**Hypothesis valid**
3	Dynol gun spray angle not ok	Visual checking	**Hypothesis valid**

5.3 Identify the root causes:

WHY - WHY Analysis

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Associate not able to protect overspray/ No control	Due to Manual operation while dynol application its get overspray on surrounding parts	Critical gap between dynol application area & body parts (less gap)	No provision for parts to prevent for Dynol overspray	Protect surrounding parts
2	No adequate sequence define while dynol application	Associate to associate variation for dynol application	Application Sequence not defined	In adequate dynol application seq.	Change the sequence
3	Dynol gun spray angle not ok	To reduce process time this failure not consider in initial level	Due to wider angle overspray happening on surrounding parts.	Current dynol spray angle is inappropriate	Change Gun Spray angle

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
1	No provision for parts to prevent for Dynol overspray	Protect surrounding parts – New FRP fixtures developed to protect from Dynol overspray.	12.7.18	Pravin Patil	Closed

Before

After

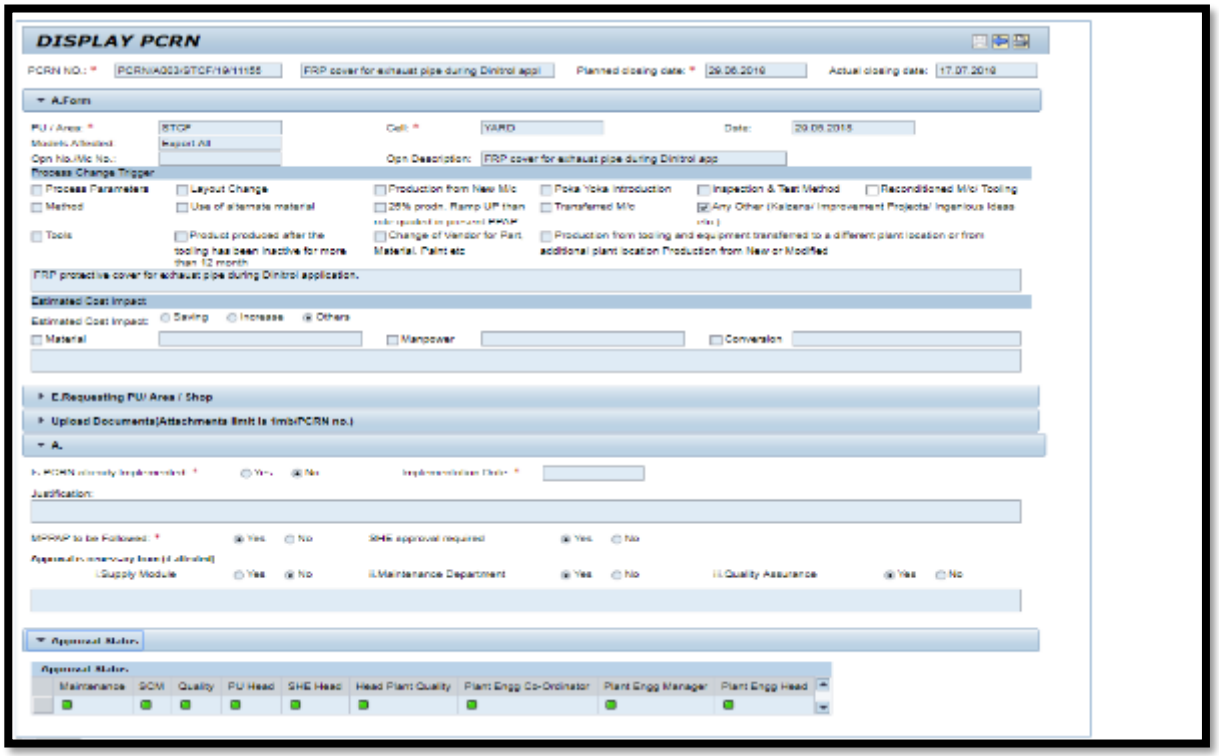
No Surrounding Protection Fixture while Dynol application

Fixture Implemented for Dynol application for critical points & Numbering given to fixtures

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
1	No provision for parts to prevent for Dynol overspray	Protect surrounding parts – New FRP fixtures developed to protect from Dynol overspray.	12.7.18	Pravin Patil	Closed

PCRN



6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
2	In adequate dynol application seq.	Change the sequence	3.6.18	Chetan Waje	Closed

Before

Dynol Application parts coverage mentioned in SOS but sequence of Dynol application not mentioned in SOS

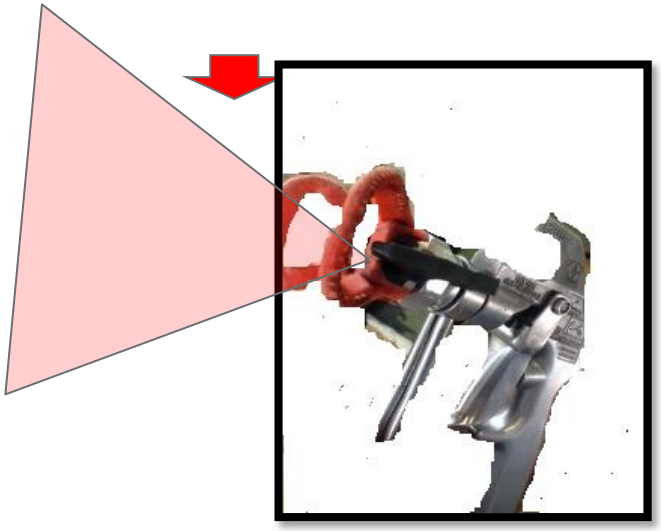
After

Dynol application sequence define in SOS

Pictorial Sequence Display On wall at stage

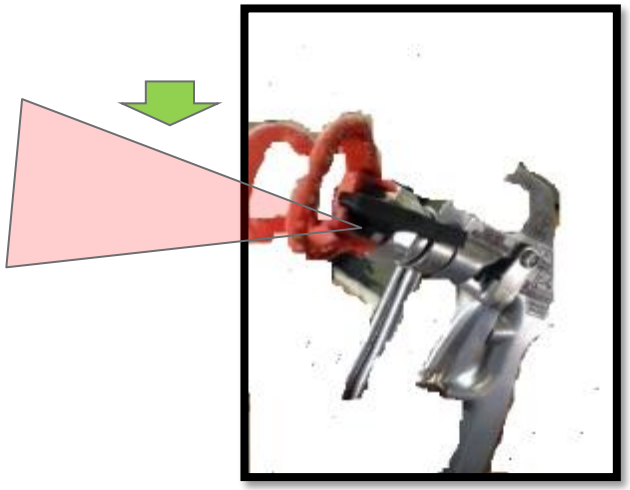
6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
3	Current dynol spray angle is inappropriate	To prevent the dynol overspray on outer A class area of vehicle --- reduced spray gun orifice nozzle size (needle 520 to needle 350).	28.6.18	Datta Jondhale	Closed



WIDER SPARY -520 needle size

Before



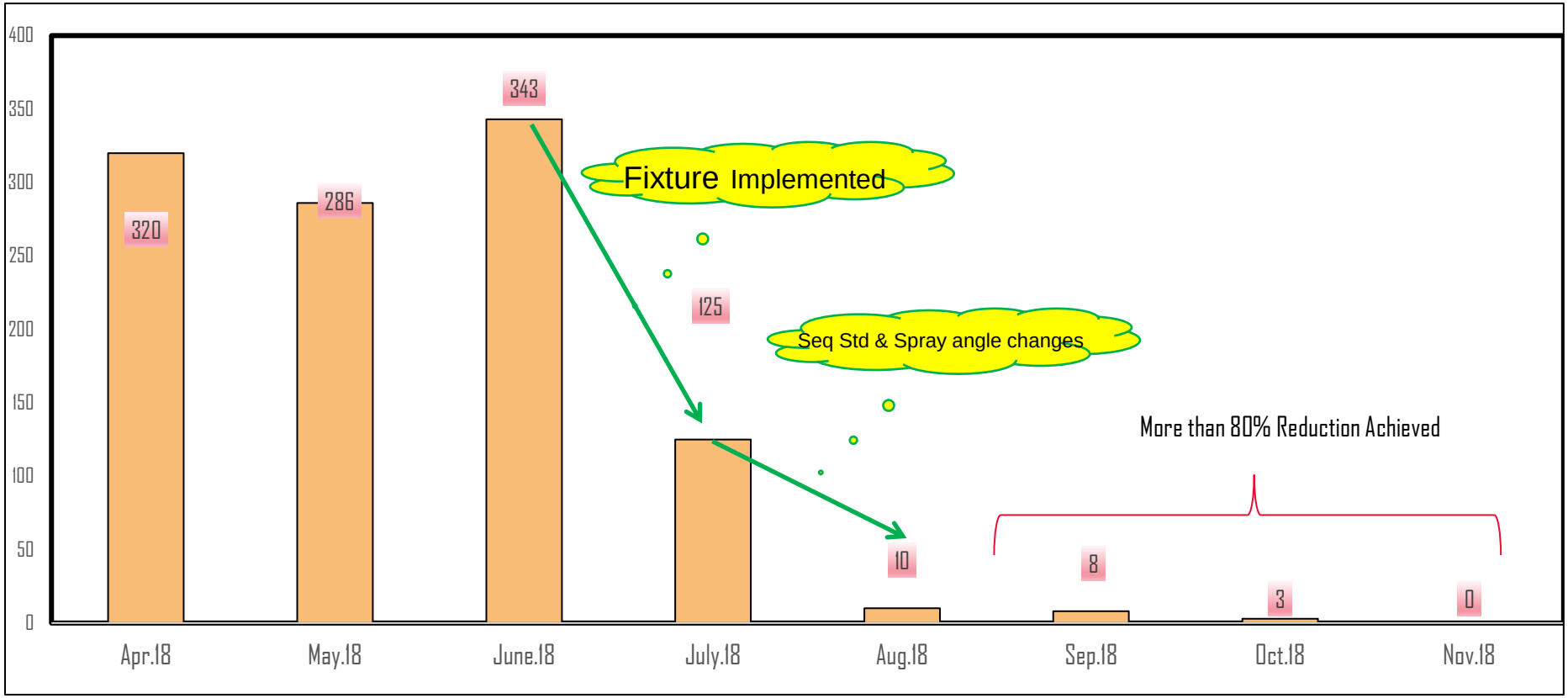
Less WIDER SPARY -350 needle size

After

7. Check & confirm the results & effects

7.1 Verify Results: Defect Trend

X Axis- Month
Y Axis-Roof liner ding at rear side in Scorpio vehicle concern numbers.



Inference - Concern is eliminated after all 2 actions implemented.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	PCRN Updated	17.7.18	Sayali more	CLOSED
2	SOS updated	3.6.18	Chetan Waje	CLOSED
3	PM check sheet updated	3.6.18	Sagar Kekade	CLOSED
4	Work content signoff for additional fixture activity	20.7.18	Rajesh Dhanokar	CLOSED
5	FRP Fixture list updated	12.7.18	Pravin Patil	CLOSED

Check & confirm the results & effects (Cont..)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) More than 80% Reduction in Dynol overspray concern.
2) RTR Improved from 89 to 91%
3) COPQ (Material cost+ Labour cost+ Rework cost) =(Apprx. Rs.50152 Rs. Per annum) is Saved
4) Repeat concern of RFI / Field resolved

In tangible Benefits

1) Knowledge of Worker is Improved.
2) Improved Customer Satisfaction.

Certified by Industrial Engineer

From: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Date: 17 December 2018 at 18:45:48 IST
To: DHANOKAR SUDHIR <DHANOKAR.SUDHIR@mahindra.com>
Subject: RE: COPQ cost for MYB

Okay Approved

From: DHANOKAR SUDHIR
Sent: Monday, December 17, 2018 5:55 PM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Subject: COPQ cost for MYB

Dear Rajesh,

Please approve COPQ cost for MYB

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	18
Total Required Rework time per day	Min	180
Effective working time	Min	465
Operator required /day	nos	0.39
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	125.81
Avg no. of working days	days	24
Operator cost /month	Rs	3019.35
Part reject/month	Nos	0.00
Part Cost	Rs	0.00
Operator Cost/year	**Rs**	**36232.26**

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	PCRN Updated	17.7.18	Sayali more	CLOSED
2	SOS updated	3.6.18	Chetan Waje	CLOSED
3	PM check sheet updated	3.6.18	Sagar Kekade	CLOSED
4	Work content signoff for additional fixture activity	20.7.18	Rajesh Dhanokar	CLOSED
5	FRP Fixture list updated	12.7.18	Pravin Patil	CLOSED

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	One point lesson to Stage associate for usage of FRP Fixture	12.7.18	Chetan Waje	Closed

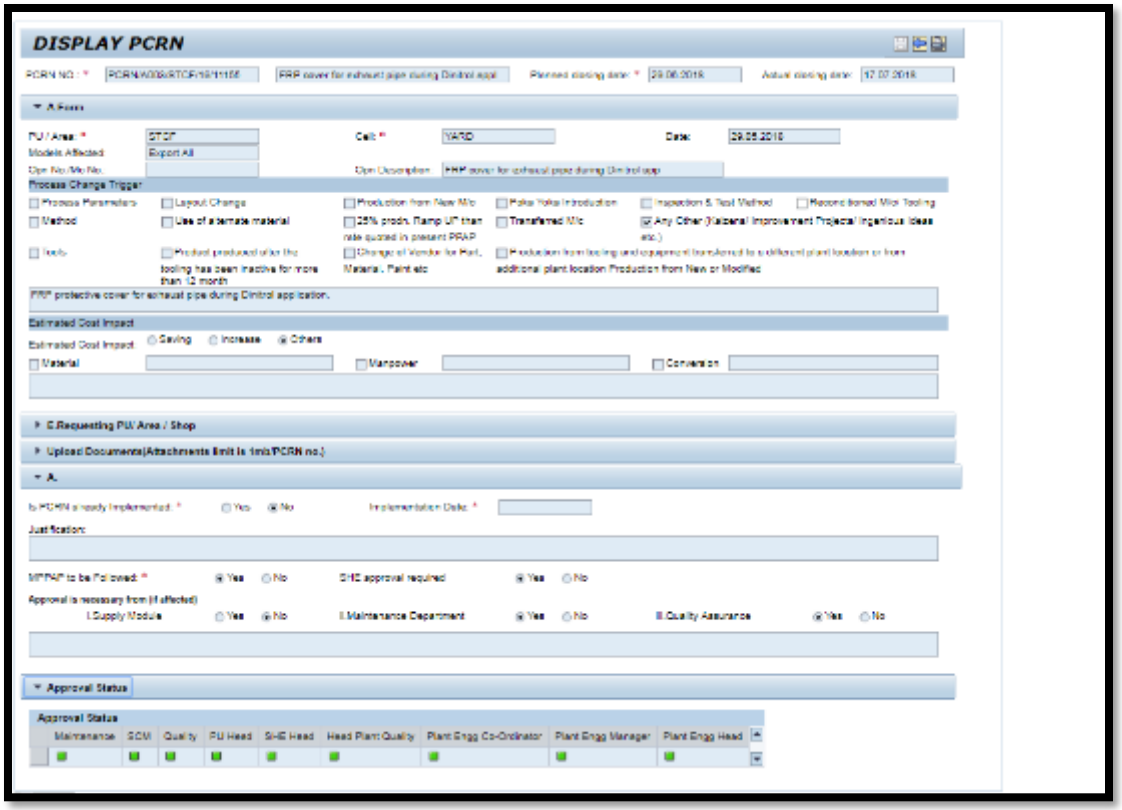
8.3 Additional Audit & check

1	Point added in supervisor check sheet	15.7.18	Chetan Waje	Closed

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

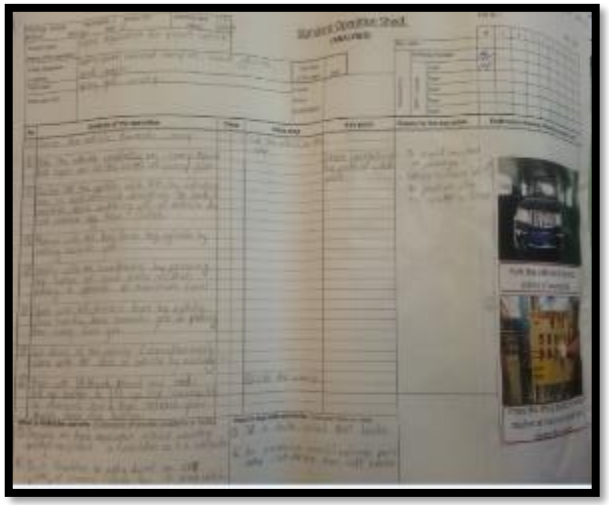
SN	Document	Actual date	Responsibility	Status
1	PCRN for Dynol fixture	17.7.18	Sayali More	CLOSED

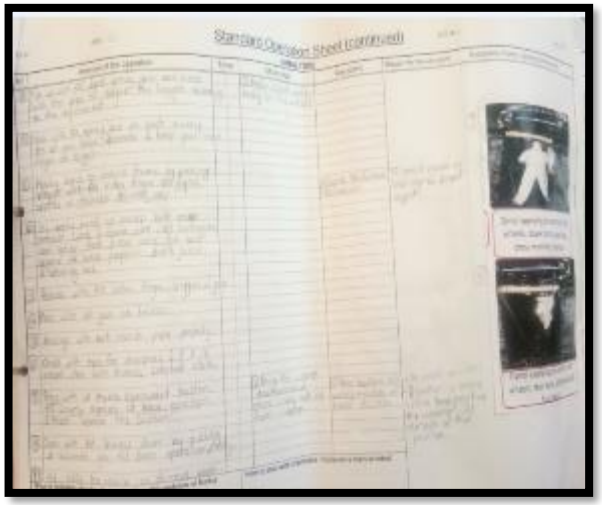


8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
2	SOS sequence modified & 100% associate training done	3.6.18	Chetan Waje	CLOSED





SOS Updated

Pictorial Sequence Display On wall at stage

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
4	Work content signoff for additional fixture activity	20.7.18	Rajesh Dhanokar	CLOSED

From: DHANOKAR RAJESH
Sent: Tuesday, December 18, 2018 1:42 PM
To: DHANOKAR SUDHIR <DHANOKAR.SUDHIR@mahindra.com>
Subject: Sent from Snipping Tool

Sr No	Operation Description	Workcontent in man min @ 5MPH												
		Scorpio	U321	MDI / LCCR	Xylo	Quanto	Quanto U215	BMT MDI	BMT LCCR	Export	Max Model Work Content	Manpower Required	Manpower Deployementt	Utilsation % as per Work content
	DYNOL APPLICATION													
1	DYNOL APPLICATION ACTIVITY	3.82		3.82	3.82	3.82	3.82	3.82	3.82	3.82				
	Vehicle Loading and unloading	1.33		1.33	1.33	1.33	1.33	1.33	1.33	1.33				
	Total	**5.15**		**5.15**	**5.15**	**5.15**	**5.15**	**5.15**	**5.15**	**5.15**	**5.15**			
	Set up : - 75										**5.15**	**0.83**	**1**	**82.82%**

Work content Dynol stage with fixture activity

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
5	FRP Fixture list updated	12.7.18	Pravin Patil	CLOSED

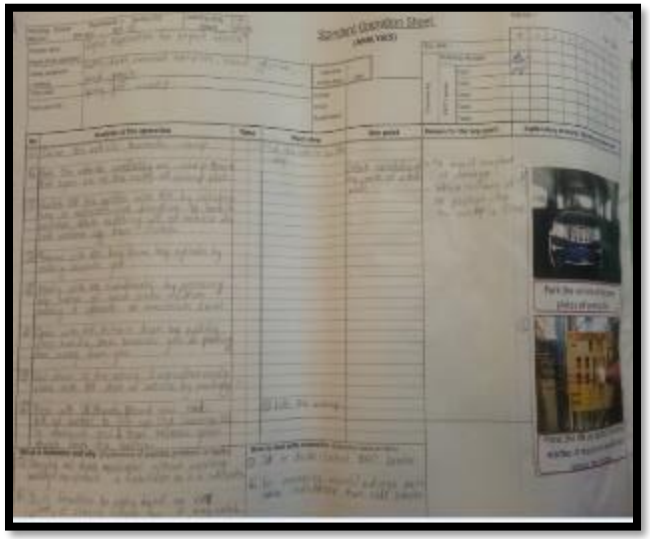
M & M Ltd. Automotive sector Nashik		Quality Management Systems TITLE : List of Fixtures			PU : SCORPIO/XYLO TCF Page : See Footer Revision No. : 4.0 Effective Date : 01-Sept-17	
Sr No.	FIXTURE DESCRIPTION	Maintenance Freq	Fixture Type	Number	Rev No.	Area
1	Door Key Cylinder Switch Sub Assy Fixture	2 Month	Sub Assy	MS-3390	00	Trim 1
2	Registration Plate lamp Mtg. Fixture	2 Months	Sub Assy	MS-3033	00	Trim 1
3	Garnish plate camera sub assy FRP-Scorpio	2 Months	Gap gauge	MS-F-026	00	Trim-1
4	Steering Column Sub Assy Fixture (Scorpio)	2 Month	Sub Assy	MS-2908	01	Trim 1
5	Steering Column Fixture Sub Assy Fixture (Xylo)	2 Month	Sub Assy	MS-3375	00	Trim 1
6	Steering column Sub Assy Fixture (Scorpio)	2 Months	Sub-assy	MS-4227	00	Trim-1
7	ECU Sub Assy Fixture -C39-Scorpio RHD/LHD	2 Month	Sub Assy	MS-2907	00	Trim 1
8	ECU Sub Assy Fixture (Xylo/Quanto)	2 Month	Sub Assy	MS-3407	00	Trim 1
9	ECU Sub Assy Fixture - C-S5-Scorpio RHD/LHD	2 Month	Sub Assy	MS-3796	00	Trim 1
10	ECU Sub Assy Fixture (Scorpio LCCR)	2 Month	Sub Assy	MS-3797	00	Trim 1
11	ECU sub-assy fixture -C55	2 Months	Sub-assy	MS-4258	00	Trim 1
12	ECU sub-assy fixture -LCCR (W110)	2 Months	Allignment	MS-4257	00	Trim 1
13	ECU sub-assy fixture - (W109)	2 Months	Allignment	TDR-F17-186	00	Trim 1
14	Fuel Neck Assy Fixture -Scorpio	2 Months	Sub Assy	MS-3035	01	Trim 1
15	Brake switch gap gauge spoon type -S/X/Q	2 Months	Locating Gauge	MS-Gauge-01	00	Trim-1
16	A-B-C gap checking gauge-S/X/Q	2 Months	Gap gauge	MS-Gauge-02	00	Trim-1
17	ABS EHCU Sub Assy Fixture (Scorpio RHD-LHD)	2 Month	Sub Assy	MS-3395	01	Trim 1
18	ABS EHCU Sub Assy Fixture(Xylo RHD/LHD)	2 Month	Sub Assy	MS-3581	00	Trim 1
19	Horn Gap Setting Gauge-Xylo /Quanto	2 Months	Allignment	MS-3386	01	Trim 1
20	SC/DC horn gap setting & alignment fixture	2 Months	Allignment	MS-F17-01	00	Trim-1
21	Rear bumper to Qtr end bumper gap setting fixture-Scorpio	2 Months	Allignment	MS-4675	00	Trim-1
22	AC pipe Assy Fixture-Xylo	2 Months	Sub Assy	MS-3403	01	Trim 1
23	Hood Opening Switch Gauge -W103	2 Months	Allignment	MS-3389	00	Trim 2
24	Hood Opening Switch Gauge Scorpio	2 Months	Locating	MS-4280	00	Trim 2
25	Qtr Glass Sub Assy Fixture (LH & RH)	2 Months	Sub Assy	MS-2911	00	Trim 2
26	Hand Brake Sub Assy Cable Mtg Fixture-Scorpio	2 Months	Sub Assy	MS-2912	00	Trim 2
27	Jump Seat Fixture (RH)-Scorpio 9 seater	2 Months	Locating	MS-3405	00	Trim 2
28	Jump Seat Fixture (LH)-Scorpio 9 seater	2 Months	Locating	MS-3406	00	Trim 2
29	Seat Latch Assy Fixture (LH& RH)-Scorpio	2 Months	Locating	MS-3029	00	Trim 2
30	Export Glass Preparation rotating stand-SC/DC	2 Months	Preparation	MS-3382	00	Trim 2
31	Butterfly Fixture (LH & RH) -Scorpio	2 Months	Locating	MS-3036	01	Trim-2
32	Butterfly Fixture (LH & RH) -Xylo	2 Months	Locating	MS-3036	00	Trim-2
33	Butterfly Fixture (LH & RH) Quanto	2 Months	Sub-assy	12F048	00	Trim-2
34	Scuffplate rocker panel protective tef. Plate- Front Lh/Rh Scorpio	2 Months	Protecting	MS-F18-01 A	00	Trim-2
35	Scuffplate rocker panel protective tef. Plate- Rear Lh/Rh Scorpio	2 Months	Protecting	MS-F18-01 B	00	Trim-2
36	Driver Arm Rest Sub Assy Fixture (LH & RH)-Xylo	2 Months	Sub Assy	MS-3381	00	Trim 3
37	Front Rh door power window console sub-assy	2 Months	Sub-assy	MS-4281	00	Trim-3
38	Pressure Modulator Sub Assy Fixture (3)-Scorpio	2 Months	Sub Assy	MS-3392	00	Trim 3
39	W/S (Ft.) Glass Preparation Rotating Stand-All model	2 Months	Preparation	MS-2342	00	Trim 3
40	Backdoor Glass Preparation Fixture-All model	2 Months	Preparation	MS-3369	00	Trim 3
41	Rear Door Glass Locating Fixture (Scorpio)	2 Months	Locating	MS-2916	00	Trim 3
42	Rear Door Glass Locating Fixture (Xylo)	2 Months	Locating	MS-3327	00	Trim 3
43	Rear door Glass Locating FRP fixture W110	2 Months	Locating	MS-4282	00	Trim 3
44	Glass rear door locatinhg fixtue Quanto	2 Months	Sub-assy	MS-3911	00	Trim 3
45	Radiator Sub Assy Fixture (all model)	2 Months	Sub Assy	MS-3367	01	Trim 3
46	FATC control panel sub assy FRP fixt.	2 Months	Allignment	MS-4283	01	Trim 3
47	RTLS cover Fitment Fixture (Scorpio)	2 Months	Allignment	MS-3802	01	Trim 3
48	Cargo Rear Bumper Sub Assy Fixture	2 Months	Sub Assy	MS-2976	00	FAI
49	Fixture For Cargo Rear Bumper Alignment (Pit)-Rh/Lh	2 Months	Allignment	MS-2972	01	FAI
50	Rear Bumper Alignment Fixture (W103)	2 Months	Allignment	MS-3404	00	FAI
51	Rear Bumper Alignment Gauge (W110/W105)	2 Months	Allignment	MS-4284	01	FAI
52	Fuel Filter Sub Assy Fixture-W103/Xylo/Quanto	2 Months	Sub Assy	MS-2994	00	FAI
53	Frnt Bumper Alignment Fixture (Scorpio)	2 Months	Alignment	MS-3069	00	FAI
54	LSPV Setting Gauge (SC/DC)	2 Months	Allignment	MS-3396	00	FAI
55	LSPV Setting gauge (Scorpio)	2 Months	Allignment	MS-2995	00	FAI
56	Ride-height checking gauge Common	2 Months	Allignment	MS-3040	00	FAI
57	Rear Axle Sub Assy Fixture common for SC/DC,BMT,Tuner subassy	2 Months	Sub Assy	MS-2928	01	Chassis
58	Caster Camber Bolt Setting Tool (Chassis)	2 Months	Allignment	MS-4485	00	Chassis
59	T-Hose Sub Assy Fixture	2 Months	Sub Assy	MS-2943	00	Chassis
60	Propeller Shaft Central bush Bearing Assy Fixture (XYLO/U215)	2 Months	Locating	MS-3370	00	Chassis
61	Propeller Shaft Central bush Bearing Assy Fixture (Old Quanto)	2 Months	Assembly	MS-4112	00	Chassis
62	Auto Transmission fixture A\T (SCORPIO)	2 Months	Locating	MS-3658	00	Trim 1

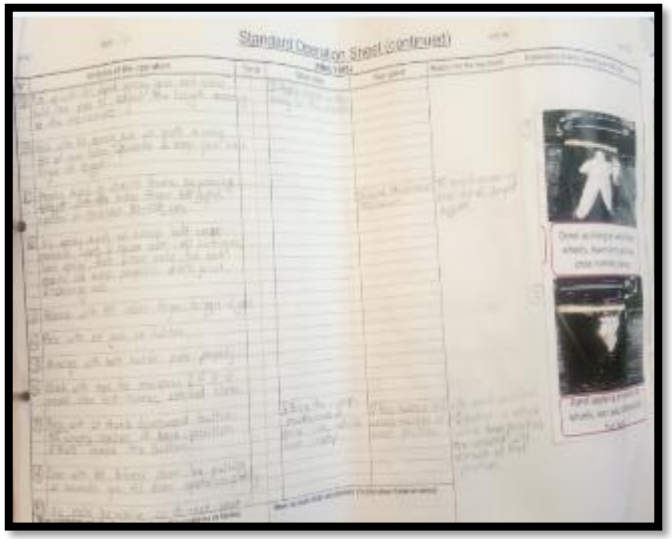
Dynol fixtures list updated

8. Standardization

8.2 Training:

Action Plan	Target Dt	Resp.	Status
Stage associate training done with modified SOS for usage of FRP Fixture	12.7.18	Chetan Waje	Closed



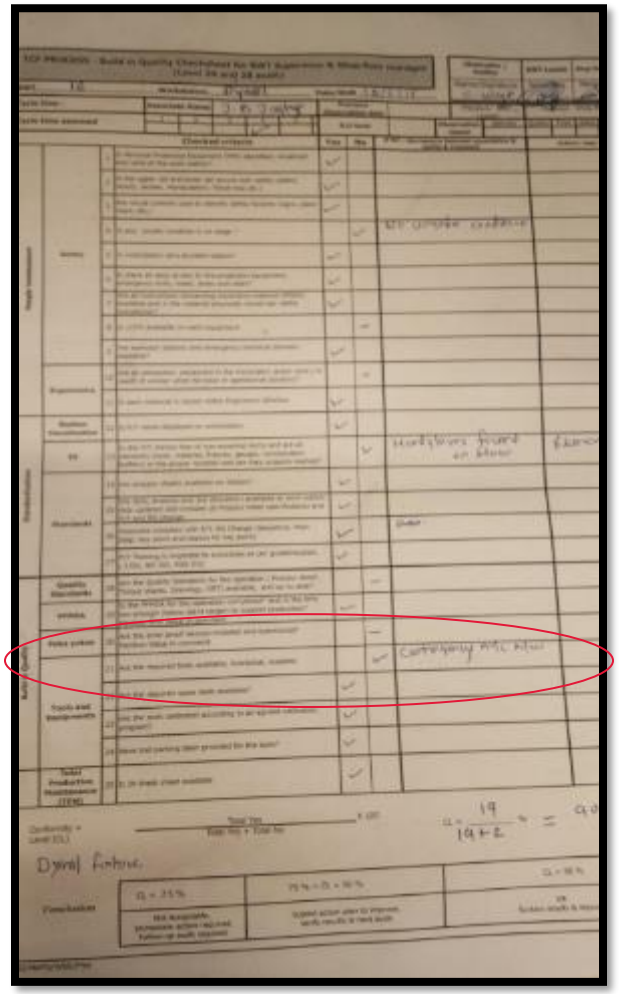


8. Standardization

8.3 Additional Audit & check

check sheet updated

Point added in supervisor check sheet- BIQ.



9. Summarise & Plan future actions

9.1 Conclusion:

After implementation of fixture, gun spray angle modification & sequence of dynol application change elimination resolved the Dynol overspray on body concern.

9.2 Horizontal Deployment:

Upcoming export models.

9.3 Lessons Learnt:

Things Gone Wrong : Dynol gun spray angle major role for overspray was not identified.

Things Gone Right :

Protecting surround parts while dynol over spraying is protected with small fixtures which are easy to fit & remove.

Thank you

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